1.4 Atomic Absorption Spectroscopy

Problems Worksheet

1. Flame tests can be used to identify elements in a chemistry lab.
   a. Using a diagram, explain how colours are produced when metal elements are heated in a Bunsen burner.

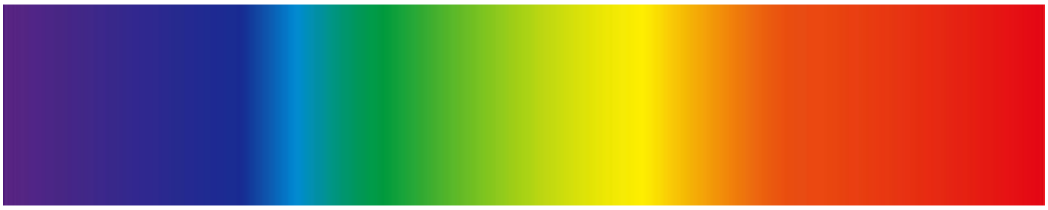
   b. How can these colours be used to identify elements?

   c. Why is this technique only useful for identifying metals?

2. The diagram below shows the emission spectrum for hydrogen.

a. All of the lines visible here are produced when electrons fall to n = 2. Explain why there are no lines produced when an electron falls back to the ground state (n = 1).

b. A continuous spectrum is shown below. Add to the diagram to show how it would appear if the light was transmitted through a sample of hydrogen gas.



c. What is the relationship between the emission and absorption spectra?

3. Describe what occurs in each of these parts of an atomic absorption spectrometer and why it is important in the process.
    a. Hollow cathode lamp

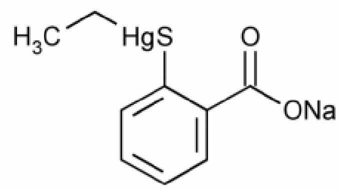
    b. Flame

    c. Nebuliser

    d. Monochromator

    e. Detector

4. Thiomersal is used as a preservative in some injectable vaccines. Preservatives have been considered extremely important in vaccines since more than half the children injected with a diphtheria vaccine in 1982 were infected with Staphylococcus and subsequently died from the infection. Effective preservatives are a challenge as many will affect the vaccine.



Thiomersal contains mercury, which is toxic in large amounts. In Australia, the maximum allowable concentration of Thiomersal in a vaccine is 0.01%, which is equivalent to 50 µg of mercury in a 1 mL dose.

Cooper and Jordan are analysing the amount of mercury in a sample of influenza vaccine using AAS. They carefully create a set of standard solutions and measure the absorbance in an atomic absorption spectrometer, using a mercury lamp. They then measure the absorbance of the vaccine. The results are shown in the table.

Concentration (µg.mL^{-1})	**Absorbance**
10	31
20	62
30	93
40	125
50	156

a. Plot the data on a calibration curve.

b. The absorbance measured for the vaccine was 48. Use the curve to determine the concentration of Hg in the vaccine sample.

c. Explain why other substances in the vaccine will not affect the results.

d. AAS is only useful for analysing metals. Explain why this is so.

